Statistics in Society

Concept

Statistics describe data. Good conclusions depend on how the data were collected.

Population / sample

Whole group of interest / subset actually studied.

Sampling

Random and stratified methods can reduce selection bias.

Bias

Check wording, non-response, timing, missing groups and graph scales.

Ethics

Protect privacy; correlation alone does not establish causation.

Example 1

A school surveys only students at the bus stop about transit use. Evaluate it.

Solution

1. Problem: convenience sample overrepresents transit users.
2. Improve: randomly select students from the whole school or within grades.

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Example 2

Of 200 invited students, 120 respond and 72 prefer A. Interpret the result.

Solution

1. Among respondents: 72/120 = 60% prefer A.
2. The other 80 did not respond; whole-school support remains uncertain.
3. Do not present 60% as a guaranteed population result.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Identify the population and sample in a survey of 60 randomly selected Grade 9 students from one school.

2. Rewrite: 'Don't you agree our amazing new timetable is better?' neutrally.

3. Does an association between ice-cream sales and swimming accidents prove ice cream causes accidents?

4. Give two checks before sharing a chart online.

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Practice answers | check after trying

1. Population: all Grade 9 students at that school; sample: the selected 60.

2. 'Compared with the previous timetable, how do you rate the new timetable?' Include balanced options and 'no opinion'.

3. No. Hot weather can increase both swimming and ice-cream purchases; association does not establish causation.

4. Check the source and date, axis scale, units, sample method and size, missing groups, and whether the conclusion fits the data. Any two with an explanation are suitable.